

Case Study Analysis

1. Case Study Overview

TDM Insight is a native Android application designed for Therapeutic Drug Monitoring (TDM) calculations. The case study is based on a hospital pharmacy department where TDM calculations may require different types of patient, medication, concentration, sampling time, and laboratory information.

The application aims to bring these inputs and calculations together in one structured mobile application. The main focus is Vancomycin TDM, with three calculation workflows:

Vancomycin Pre, Vancomycin Post, and Vancomycin Pre + Post.

The project uses Kotlin and Android Studio with Jetpack Compose or XML for the user interface and Material 3 for the application design. The main challenge is not only to create a calculator screen, but to convert a non-trivial TDM calculation workflow into a reliable and understandable mobile application.

2. Problem Analysis

The main problem identified in the case study is the difficulty of handling a non-trivial Therapeutic Drug Monitoring calculation workflow in a clear and reliable way. TDM calculations may require several types of information, including patient parameters, medication dose, dosing interval, drug concentration, sampling time, and laboratory information.

Another challenge is that different Vancomycin calculation methods require different inputs.

The application needs to support Pre, Post, and Pre + Post workflows, and each workflow should display only the input fields required for that method. This can make the input process more organized and reduce unnecessary information on the screen.

Input validation is also an important issue. The application should check required fields, numeric values, units, specified ranges, and logical relationships between fields. It should also prevent mathematical problems such as division by zero or invalid logarithmic operations.

Finally, the calculation results should be understandable to the user. Instead of displaying only the final result, the application should provide a simplified explanation showing the important input values, intermediate calculations, pharmacokinetic parameters, and final result.

Therefore, the proposed application should focus on three main areas: collecting the correct information, performing reliable calculations, and presenting the results in an understandable way.

3. Target Users

The main target users of TDM Insight are hospital pharmacy staff who perform Therapeutic Drug Monitoring (TDM) calculations. The application is designed to support their calculation workflow by providing a structured way to enter the required information, validate the inputs, perform the calculation, and review the results.

The application is intended as an academic software prototype. It is not designed to replace professional clinical judgement or act as an autonomous treatment-decision system.

4. Project Objectives

4.1 Main Objective

The main objective of TDM Insight is to develop a reliable native Android TDM Calculator that can support Vancomycin Therapeutic Drug Monitoring calculations through a clear and structured mobile application.

4.2 Specific Objectives

1. Develop the application using Kotlin and Android Studio with Jetpack Compose or XML.
2. Implement the three required Vancomycin TDM workflows: Pre, Post, and Pre + Post.
3. Provide dynamic input forms that display the required fields according to the selected calculation method.
4. Validate user inputs, including required fields, numeric values, units, ranges, and logical relationships between related fields.
5. Separate the TDM calculation engine from the user interface to improve the structure of the application.
6. Display relevant intermediate and final pharmacokinetic results.
7. Provide a simple calculation explanation so that users can understand how the main results were obtained.
8. Provide a clean and intuitive user interface following Material 3 design principles.

5. Functional Requirements

The functional requirements describe what the TDM Insight application should allow the user to do.

ID	Requirement	Description
FR1	Create Fictional Case	The user should be able to create a fictional TDM case for demonstration.
FR2	Enter Patient Parameters	The application should allow the user to enter the required patient parameters.
FR3	Select Vancomycin	The user should be able to select Vancomycin as the medication.
FR4	Select TDM Workflow	The user should be able to select Pre, Post, or Pre + Post calculation.
FR5	Dynamic Input Fields	The application should display input fields according to the selected TDM workflow.
FR6	Validate Inputs	The application should validate required fields, numeric values, units, ranges, and relevant relationships between inputs.
FR7	Run Calculation	The application should process the validated inputs using the TDM calculation engine.
FR8	Display Intermediate Results	The application should display relevant intermediate pharmacokinetic values.
FR9	Display Final Results	The application should display the final pharmacokinetic results produced by the calculation.
FR10	Explain Calculation	The application should provide a simplified explanation showing the important inputs, intermediate calculations, and final result.

6. Non-Functional Requirements

The non-functional requirements focus on the quality, reliability, usability, and structure of the application.

ID	Requirement	Description
NFR1	Usability	The application should provide a clear and intuitive user interface.
NFR2	Reliability	The calculation process should produce reliable results based on the approved calculation specifications.
NFR3	Validation	The application should prevent invalid inputs and mathematical errors where possible.
NFR4	Understandability	The results should be presented in a clear way, including an explanation of the main calculation steps.
NFR5	Maintainability	The calculation engine should be separated from the user interface to keep the application structured and easier to maintain.
NFR6	UI Design	The application should follow a clean Material 3 interface design.

7. Business Rules and Constraints

The following business rules and constraints are identified from the case study:

ID	Rule / Constraint	Description
BR1	Fictional Cases	All demonstration cases must use fictional patient information.
BR2	Workflow-Based Inputs	The selected Vancomycin workflow determines which input fields are displayed.
BR3	Valid Clinical Equations	Clinical equations, units, assumptions, and reference values must be based on lecturer-approved authoritative sources.
BR4	Input Validation	The application must validate required fields, numeric values, units, ranges, and relevant relationships between fields.
BR5	Mathematical Protection	The application should prevent mathematical errors such as division by zero and invalid logarithmic operations.
BR6	Academic Purpose	The application is an academic software prototype and must not be presented as a clinically validated or autonomous treatment-decision system.

8. Mandatory Features

Based on the case study, the application should implement the following mandatory features:

1. Native Android development using Kotlin and Android Studio.
2. A user interface developed using Jetpack Compose or XML.
3. Material 3 design principles for the user interface.
4. Vancomycin TDM calculation workflows.
5. Vancomycin Pre calculation.
6. Vancomycin Post calculation.
7. Vancomycin Pre + Post calculation.
8. Dynamic input forms based on the selected calculation workflow.
9. Required-field, numeric, unit, range, and cross-field validation where applicable.
10. A dedicated calculation engine separated from the user interface.
11. Display of relevant intermediate pharmacokinetic results.
12. Display of final calculation results.
13. A calculation explanation showing the main inputs, intermediate calculations, and final results.

9. Optional Enhancements

The case study also provides several optional features that can be added to improve the application. These features are not required for the core project.

The possible enhancements include:

1. Local calculation history.
2. Camera capture of a fictional laboratory report or medication label.
3. OCR-assisted extraction of selected values, with user confirmation.
4. What-if or scenario simulation.
5. An additional TDM medication module.
6. A simple concentration-time graph.
7. Exporting or sharing a calculation summary.

If the camera and OCR features are implemented, the extracted values must be reviewed and confirmed by the user before they are used in the calculation.

10. User Flow Analysis

The proposed user flow follows the calculation process described in the case study.

First, the user opens the TDM Insight application and creates a fictional case. The user then enters the required patient parameters and selects Vancomycin. After that, the user chooses one of the three available calculation workflows: Pre, Post, or Pre + Post.

The application then displays the input fields required for the selected workflow. After entering the required values, the application validates the information before allowing the calculation to proceed.

The user can review the calculation inputs and then run the TDM calculation. The application displays the intermediate results and final results. Finally, the user can open the calculation explanation to understand how the main results were obtained.

This flow follows the case study's main process of:

Input → Validation → Calculation → Results → Explanation

11. Technical Requirements

The TDM Insight application is required to be developed as a native Android application. The main technical requirements identified from the case study are:

Area	Requirement
Platform	Native Android
Programming Language	Kotlin
Development Environment	Android Studio
UI Framework	Jetpack Compose or XML
UI Design	Material 3
Calculation Logic	Dedicated TDM calculation engine
Application Structure	Calculation logic should be separated from UI functions
Validation	Input and cross-field validation
Results	Structured intermediate and final results

12. Proposed Solution

TDM Insight proposes a native Android solution that organizes the TDM calculation process into a clear and structured workflow. Instead of placing all possible information on one screen, the application provides different input fields according to the selected Vancomycin calculation method.

The application first collects the required information and validates the user's inputs. The validated data is then passed to a dedicated calculation engine. Separating the calculation engine from the user interface helps keep the calculation logic organized and maintainable. After the calculation is completed, the application displays the relevant intermediate and final pharmacokinetic results. Users can also open a calculation explanation to understand the main steps used to obtain the result.

Overall, the proposed solution addresses the case study by combining dynamic input, validation, calculation, and explainable results in one native Android application.

13. Conclusion

The case study presents a need for a structured and reliable mobile solution for Vancomycin Therapeutic Drug Monitoring calculations. The main challenges involve handling different input requirements, validating the information, performing non-trivial calculations, and presenting the results clearly.

TDM Insight addresses these challenges by providing three Vancomycin workflows, dynamic input forms, input validation, a separate calculation engine, and explainable results. The application will be developed as a native Android application using Kotlin and Android Studio.

The analysis shows that the project is not simply about creating a calculator. It requires the team to understand the TDM workflow and translate the requirements into a reliable and understandable mobile application. As an academic prototype, the application will use fictional cases and follow appropriate clinical references for its calculations.